

Usual Transforms in Power Systems

ELEC0447 - Analysis of electric power and energy systems

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Lecture Outline

Motivation: Why Transform in Power Systems?

Direct, Inverse, and Zero Sequence Decomposition

The Clarke Transform ($\alpha\beta 0$ Frame)

The Park Transform ($dq 0$ Frame) and Grid Synchronization

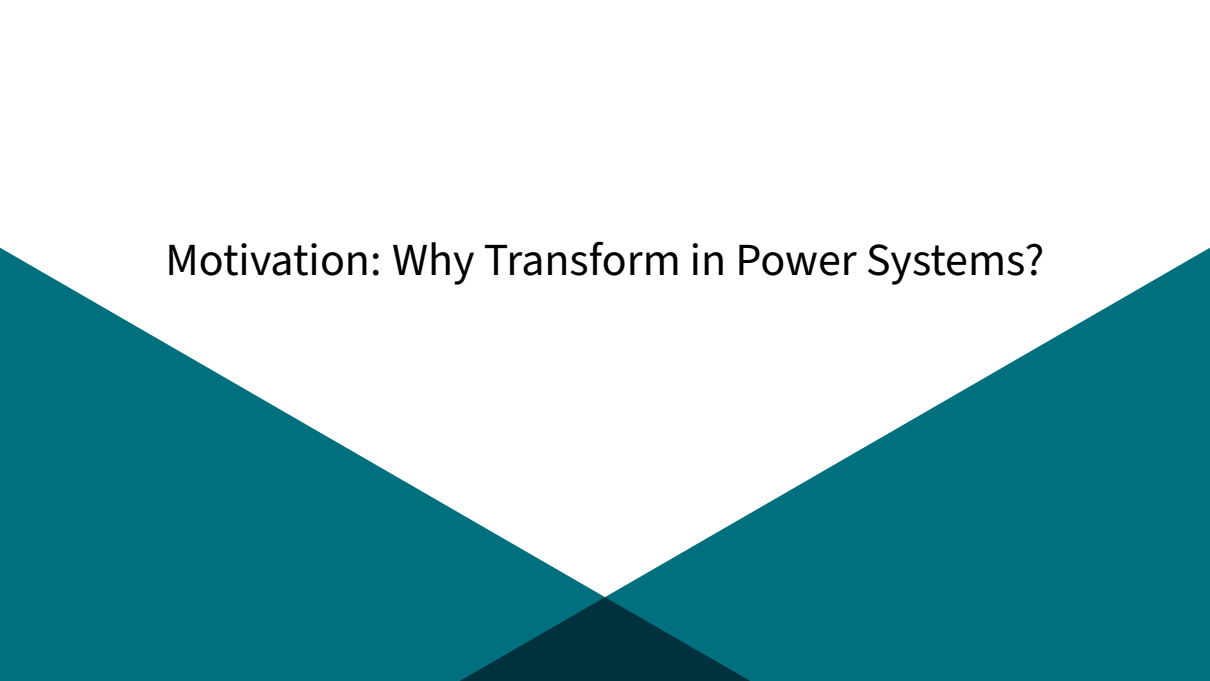
Bridges and Connections Between Transforms

Kron Reduction (Network Admittance Reduction)

Discrete Fourier Transform (DFT) for Phasor Estimation

Summary and Conclusions

Motivation: Why Transform in Power Systems?

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The Physical Reality of Power Systems I

Power systems are composed of three-phase physical networks:

- ▶ Generators with rotating magnetic fields.
- ▶ Transmission lines with mutual magnetic and capacitive coupling between phases.
- ▶ Transformers with coupled iron cores and diverse winding connections.
- ▶ Loads that can be single-phase, unbalanced, non-linear, or dynamic.

The Physical Reality of Power Systems II

The 4 Fundamental Mathematical Challenges

1. **Electromagnetic Phase Coupling:** Off-diagonal elements in $\mathbf{Z}_{abc}$ and $\mathbf{Y}_{abc}$ prevent independent per-phase analysis.
2. **Time Variance:** Sinusoidal AC steady states require tracking time-varying quantities; in rotating machines, inductances depend on rotor position $\theta(t)$.
3. **Asymmetries and Faults:** Short-circuits (single line-to-ground, line-to-line) destroy phase symmetry.
4. **High Dimensionality:** Thousands of interior buses with zero power injection burden dynamic simulations.

What Transformations Accomplish: Machines & Inverters

From AC to DC: Park Transform ($dq0$)

- ▶ "Freezes" rotating vectors onto a synchronous reference frame.
- ▶ AC quantities become constant DC quantities in steady state.
- ▶ Enables standard linear PI control for grid converters and machines.

From 3D to 2D: Clarke Transform ($\alpha\beta0$)

- ▶ Projects abc into orthogonal stationary axes.
- ▶ Foundation of space vector modulation (SVPWM) and instantaneous power theory ($p-q$).

What Transformations Accomplish: Asymmetries & Networks

Decoupling Asymmetries: Fortescue (012)

- ▶ Decomposes unbalanced 3-phase systems into 3 decoupled balanced sets.
- ▶ Diagonalizes symmetric cyclic impedance matrices.
- ▶ Essential for unsymmetrical fault analysis and protective relaying.

Dimensional Reduction: Kron Reduction

- ▶ Eliminates passive zero-injection load buses.
- ▶ Schur complement preserves exact multiport terminal behavior for stability studies.

Taxonomy of Power System Transforms

Transformation	Domain	Primary Operation	Key Power System Use Case
Fortescue	Phasor / Freq.	Change of complex basis	Unbalanced faults, sequence networks, protection relays (46, 50N/51N)
Clarke	Time / Instantaneous	3-phase to 2-axis stationary ($\alpha\beta 0$)	Space Vector PWM, instantaneous power theory ($p-q$), fast unbalance detection
Park	Time / Instantaneous	2-axis rotation to synchronous frame ($dq0$)	Machine modeling, vector control of grid converters (VSC/HVDC), SRF-PLL
Kron Reduction	Admittance Network	Multiport matrix elimination (Schur complement)	Transient stability dynamic simulation, Ward equivalents, bus reduction
Full-Cycle DFT	Discrete Time to Phasor	Discrete harmonic filtering	Digital distance protection, PMU synchrophasors (IEEE C37.118)

Table 1: Landscape of common power system transformations.

Direct, Inverse, and Zero Sequence Decomposition

The background of the slide features a white upper section and a teal lower section. The teal section is composed of two large triangles that meet at a point at the bottom center, creating a V-shape. The left triangle points downwards and to the right, while the right triangle points downwards and to the left.

The Symmetrical Components (Fortescue, 1918) I

Charles LeGeyt Fortescue published in 1918:

"Method of Symmetrical Co-ordinates Applied to the Solution of Polyphase Networks"

Fundamental Theorem: Any set of N unbalanced phasors can be decomposed into N symmetrical (balanced) sets of phasors of different phase sequences.

The Symmetrical Components (Fortescue, 1918) II

For a 3-phase system ($N = 3$), any unbalanced voltages $\bar{V}_a, \bar{V}_b, \bar{V}_c$ or currents $\bar{I}_a, \bar{I}_b, \bar{I}_c$ are uniquely decomposed into:

1. **Positive (Direct) Sequence (1 or +):** Three phasors of equal magnitude, displaced by 120° , with sequence $a \rightarrow b \rightarrow c$.
2. **Negative (Inverse) Sequence (2 or -):** Three phasors of equal magnitude, displaced by 120° , with reversed sequence $a \rightarrow c \rightarrow b$.
3. **Zero Sequence (0):** Three identical phasors of equal magnitude and identical phase angle.

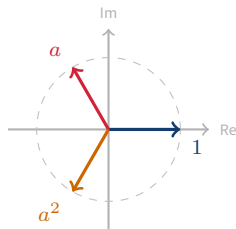
The Complex Operator a

The complex phase-shift operator a rotates a phasor by 120° counter-clockwise:

$$a = e^{j\frac{2\pi}{3}} = -\frac{1}{2} + j\frac{\sqrt{3}}{2} = 1\angle 120^\circ \quad (1)$$

Key Algebraic Properties:

- ▶ $a^2 = e^{j\frac{4\pi}{3}} = -\frac{1}{2} - j\frac{\sqrt{3}}{2} = 1\angle -120^\circ$
- ▶ $a^3 = 1, \quad a^4 = a$
- ▶ $1 + a + a^2 = 0$
- ▶ $1 - a = \sqrt{3}e^{-j30^\circ}$
- ▶ $1 - a^2 = \sqrt{3}e^{j30^\circ}$



Mathematical Definition: Synthesis and Analysis I

Expressing phase phasors from sequence components (**Synthesis**):

$$\bar{V}_a = \bar{V}_0 + \bar{V}_1 + \bar{V}_2 \quad (2)$$

$$\bar{V}_b = \bar{V}_0 + a^2 \bar{V}_1 + a \bar{V}_2 \quad (3)$$

$$\bar{V}_c = \bar{V}_0 + a \bar{V}_1 + a^2 \bar{V}_2 \quad (4)$$

In matrix form:

$$\underbrace{\begin{bmatrix} \bar{V}_a \\ \bar{V}_b \\ \bar{V}_c \end{bmatrix}}_{\mathbf{V}_{abc}} = \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} \bar{V}_0 \\ \bar{V}_1 \\ \bar{V}_2 \end{bmatrix}}_{\mathbf{V}_{012}} \quad (5)$$

Mathematical Definition: Synthesis and Analysis II

Inverting matrix $\mathbf{T}$ yields the **Analysis** transformation:

$$\mathbf{V}_{012} = \mathbf{T}^{-1} \mathbf{V}_{abc} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \bar{V}_a \\ \bar{V}_b \\ \bar{V}_c \end{bmatrix} \quad (6)$$

Mathematical Definition: Synthesis and Analysis III

Explicitly:

$$\bar{V}_0 = \frac{1}{3}(\bar{V}_a + \bar{V}_b + \bar{V}_c) \quad \textbf{(Average phasor / ground unbalance)} \quad (7)$$

$$\bar{V}_1 = \frac{1}{3}(\bar{V}_a + a\bar{V}_b + a^2\bar{V}_c) \quad \textbf{(Direct / Positive component)} \quad (8)$$

$$\bar{V}_2 = \frac{1}{3}(\bar{V}_a + a^2\bar{V}_b + a\bar{V}_c) \quad \textbf{(Inverse / Negative component)} \quad (9)$$

Physical Meaning of the Three Sequences I

Positive (Direct) Sequence $\bar{V}_1, \bar{I}_1$

- ▶ Normal operating condition of the grid.
- ▶ Generates a stator magnetic field rotating **forward** at synchronous speed ω .
- ▶ Delivers useful electromagnetic torque in motors and generators.

Physical Meaning of the Three Sequences II

Negative (Inverse) Sequence $\bar{V}_2, \bar{I}_2$

- ▶ Caused by load unbalance, single-phase faults, or untransposed lines.
- ▶ Generates a magnetic field rotating in the **reverse direction** ($-\omega$).
- ▶ Relative velocity with respect to the rotor is 2ω (100 Hz / 120 Hz).
- ▶ Induces dangerous double-frequency eddy currents in rotor surface and damper windings → **severe thermal overheating** and pulsating torque.

Physical Meaning of the Three Sequences III

Zero Sequence $\bar{V}_0, \bar{I}_0$

- ▶ All 3 phase currents I_0 oscillate completely in phase.
- ▶ Neutral / earth current: $I_n = I_a + I_b + I_c = 3I_0$.
- ▶ **Can only circulate if a neutral connection or ground return path exists!**
- ▶ Generates a pulsating, non-rotating magnetic field in the airgap.

Decoupling of Symmetrical Networks I

Consider a fully balanced three-phase line or equipment with self impedance Z_s and mutual impedance Z_m :

$$\mathbf{Z}_{abc} = \begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix} \quad (10)$$

In phase coordinates, phase a voltage drop depends on currents in phases b and c !

Applying the Fortescue transformation $\mathbf{Z}_{012} = \mathbf{T}^{-1}\mathbf{Z}_{abc}\mathbf{T}$:

$$\mathbf{Z}_{012} = \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \quad (11)$$

Decoupling of Symmetrical Networks II

where:

$$Z_1 = Z_s - Z_m \quad \textbf{(Positive-sequence impedance)} \quad (12)$$

$$Z_2 = Z_s - Z_m = Z_1 \quad \textbf{(Negative-sequence impedance)} \quad (13)$$

$$Z_0 = Z_s + 2Z_m \quad \textbf{(Zero-sequence impedance)} \quad (14)$$

Decoupling of Symmetrical Networks III

The Magic of Symmetrical Components

In any cyclically symmetric equipment, **positive, negative, and zero sequence networks are completely decoupled!**

- ▶ Positive-sequence currents produce **only** positive-sequence voltage drops.
- ▶ Negative-sequence currents produce **only** negative-sequence voltage drops.
- ▶ Zero-sequence currents produce **only** zero-sequence voltage drops.

Sequence Impedances of Power System Equipment I

1. Transmission Lines:

- ▶ Transposed lines are static: $Z_1 = Z_2$.
- ▶ Zero sequence requires current return through the earth and ground wires.
- ▶ Because earth has finite conductivity and the return loop is large (Carson's formula):

$$Z_0 \approx 2 \text{ to } 3.5 \times Z_1 \quad (15)$$

2. Synchronous Machines:

- ▶ Stator is rotating relative to fields: $Z_1 \neq Z_2 \neq Z_0$!

Sequence Impedances of Power System Equipment II

- ▶ Z_1 : dynamic reactance (X_d'' , X_d' , X_d depending on time horizon).
- ▶ Z_2 : determined by damper windings seeing a 2ω backward field ($X_2 \approx \frac{X_d'' + X_q''}{2}$).
- ▶ Z_0 : very small leakage reactance ($X_0 < X_2 < X_1$).

3. Transformers:

- ▶ $Z_1 = Z_2 = Z_{sc}$ (short-circuit impedance).
- ▶ Z_0 depends heavily on the core type and **winding connections**!
 - ▶ Delta (Δ) winding: circulates zero sequence internally, but blocks it from reaching the external network!
 - ▶ Ungrounded Wye (Y): infinite zero-sequence impedance (open circuit).
 - ▶ Grounded Wye (Y_n): allows zero-sequence flow to ground.

Example 1: Numerical Decomposition of a Voltage Sag I

Suppose phase a suffers a voltage sag due to a remote fault, while phases b and c remain unaffected:

$$\bar{V}_a = 115\angle 0^\circ \text{ V}, \quad \bar{V}_b = 230\angle -120^\circ \text{ V}, \quad \bar{V}_c = 230\angle 120^\circ \text{ V} \quad (16)$$

Let us calculate the symmetrical components using $\mathbf{V}_{012} = \mathbf{T}^{-1}\mathbf{V}_{abc}$:

$$\bar{V}_0 = \frac{1}{3}(115 + 230e^{-j120^\circ} + 230e^{j120^\circ}) = \frac{115 - 230}{3} = -\mathbf{38.33}\angle 0^\circ \text{ V} \quad (17)$$

$$\bar{V}_1 = \frac{1}{3}(115 + 230e^{0^\circ} + 230e^{0^\circ}) = \frac{115 + 460}{3} = \mathbf{191.67}\angle 0^\circ \text{ V} \quad (18)$$

$$\bar{V}_2 = \frac{1}{3}(115 + 230e^{j120^\circ} + 230e^{-j120^\circ}) = \frac{115 - 230}{3} = -\mathbf{38.33}\angle 0^\circ \text{ V} \quad (19)$$

Example 1: Numerical Decomposition of a Voltage Sag II

Interpretation:

- ▶ The positive sequence dropped from 230 V to 191.67 V.
- ▶ A negative-sequence voltage of 38.33 V (20% unbalance) appeared.
- ▶ A zero-sequence voltage of 38.33 V appeared, which can be measured across the neutral point!

Example 2: Unsymmetrical Fault Analysis I

Symmetrical components allow solving unsymmetrical faults by connecting the three decoupled sequence networks at the fault point!

Single Line-to-Ground (SLG)

Phase a to ground with fault impedance Z_f :

$$I_b = 0, \quad I_c = 0, \quad V_a = Z_f I_a \quad (20)$$

Transforming to sequence currents:

$$I_0 = I_1 = I_2 = \frac{1}{3} I_a \quad (21)$$

⇒ **Networks in SERIES!**

Line-to-Line (L-L) Fault

Phases b and c shorted together:

$$I_a = 0, \quad I_b = -I_c, \quad V_b = V_c \quad (23)$$

Transforming to sequence currents:

$$I_0 = 0, \quad I_1 = -I_2 \quad (24)$$

⇒ **Networks in PARALLEL!** Zero sequence is disconnected.

Example 3: Protective Relaying Applications

Symmetrical components are directly used to configure numerical protection relays:

Negative-Sequence Overcurrent (46)

- ▶ Monitors magnitude of $|I_2|$.
- ▶ Protects generators against rotor overheating caused by unbalances.
- ▶ Tripping characteristic:

$$I_2^2 t \leq K \quad (26)$$

where K is the generator thermal capability constant.

Ground Overcurrent (50N / 51N)

- ▶ Monitors residual current
 $3I_0 = I_a + I_b + I_c$.
- ▶ In balanced conditions, $3I_0 \approx 0$.
- ▶ Any ground fault immediately injects large I_0 .
- ▶ Detects high-impedance earth faults that standard overcurrent (50/51) misses completely.

The Clarke Transform ($\alpha\beta 0$ Frame)

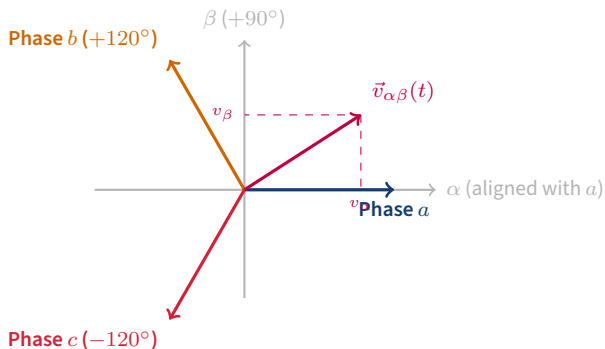
The Clarke Transform (Edith Clarke, 1938) I

Edith Clarke introduced the $\alpha\beta 0$ transformation to simplify the calculation of unsymmetrical three-phase circuits.

Core Concept:

- ▶ In a balanced three-phase system, $v_a(t) + v_b(t) + v_c(t) = 0$.
- ▶ Thus, three variables are redundant: there are only **2 degrees of freedom!**
- ▶ The Clarke transform converts the 3 stationary physical axes (a, b, c , separated by 120°) into 2 orthogonal stationary axes (α, β , separated by 90°), plus a zero-sequence axis (0).

The Clarke Transform (Edith Clarke, 1938) II



Mathematical Formulations: Amplitude vs Power Invariant I

Mathematical Formulations: Amplitude vs Power Invariant II

Amplitude-Invariant Clarke Transform (Standard in Control)

Preserves the peak amplitude of phase quantities ($\hat{V}_\alpha = \hat{V}_a$ in balanced conditions):

$$\begin{bmatrix} v_\alpha \\ v_\beta \\ v_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \quad (27)$$

Inverse Transform:

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 1 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 1 \end{bmatrix} \begin{bmatrix} v_\alpha \\ v_\beta \\ v_0 \end{bmatrix} \quad (28)$$

Power-Invariant Clarke Transform (Concordia Transform)

Uses an orthonormal matrix ($\mathbf{C}^{-1} = \mathbf{C}^T$) to preserve instantaneous power

The Space Vector Trajectory I

The complex space vector in the stationary plane is:

$$\vec{v}_{\alpha\beta}(t) = v_{\alpha}(t) + jv_{\beta}(t) \quad (30)$$

Balanced Conditions

- ▶ $v_a(t) = \hat{V} \cos(\omega t)$
- ▶ $v_{\alpha}(t) = \hat{V} \cos(\omega t)$
- ▶ $v_{\beta}(t) = \hat{V} \sin(\omega t)$
- ▶ $v_0(t) = 0$

⇒ Traces a **perfect circle of radius $\hat{V}$** rotating at angular speed ω .

Unbalanced Conditions

- ▶ Forward circle + backward circle:

$$\vec{v}(t) = \hat{V}_1 e^{j(\omega t + \phi_1)} + \hat{V}_2 e^{-j(\omega t - \phi_2)}$$

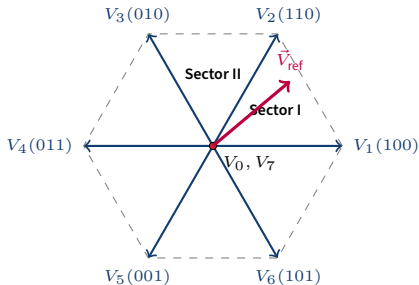
⇒ Traces an **ellipse** on the $\alpha\beta$ plane.
The ellipticity directly measures voltage unbalance!

Practical Example: Space Vector PWM (SVPWM) I

In voltage-source converters (wind, solar, battery inverters, HVDC-VSC):

- ▶ A 2-level inverter has 8 switching states ($S_a, S_b, S_c \in \{0, 1\}$).
- ▶ In the Clarke $\alpha\beta$ plane, the 8 states map to **6 active voltage vectors** forming a regular hexagon + 2 zero vectors at the origin!

Practical Example: Space Vector PWM (SVPWM) II



Result: Generating the target reference voltage $\vec{V}_{ref}$ simply requires duty-cycle calculation between adjacent sector boundary vectors.

The Park Transform ($dq0$ Frame) and Grid Synchronization

The Park Transform (Robert H. Park, 1929) I

In 1929, Robert H. Park published a landmark paper in power engineering:
"Two-reaction theory of synchronous machines"

The Fundamental Machine Problem:

- ▶ In a salient-pole synchronous generator, the air-gap magnetic permeance varies with rotor angle $\theta(t)$.
- ▶ Stator self and mutual inductances $L_{aa}(\theta)$, $L_{ab}(\theta)$ are **periodic functions of rotor position**:

$$L_{aa}(\theta) = L_s + L_m \cos(2\theta) \quad (31)$$

The Park Transform (Robert H. Park, 1929) II

- ▶ This makes machine differential equations time-varying, coupled, and non-linear!

Park's Brilliant Solution: Attach the reference frame to the **rotating rotor** (rotating at speed $\omega = \frac{d\theta}{dt}$). In this rotating dq frame, the rotor appears stationary $\implies$ **all inductances become constant!**

Mathematical Definition: Stationary to Rotating Frame I

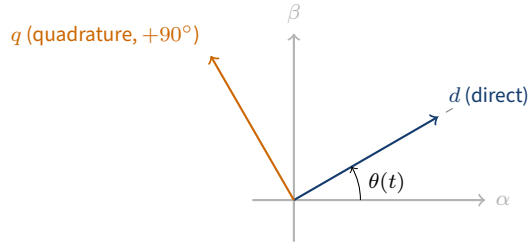
The Park transform rotates the stationary $\alpha\beta$ axes by angle $\theta(t)$:

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} \quad (32)$$

Inverse Park Transform:

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} v_d \\ v_q \end{bmatrix} \quad (33)$$

Mathematical Definition: Stationary to Rotating Frame II



Direct $abc \rightarrow dq0$ Formulation

Combining the Clarke and Park matrices yields the direct $abc \rightarrow dq0$ transformation:

$$\begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) \\ -\sin \theta & -\sin(\theta - \frac{2\pi}{3}) & -\sin(\theta + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \quad (34)$$

where $\theta(t) = \int \omega(t)dt$.

The Ultimate Magic: AC Becomes DC

For balanced three-phase sinusoids with phase angle ϕ :

$$v_a(t) = \hat{V} \cos(\omega t + \phi) \quad (35)$$

$$v_b(t) = \hat{V} \cos(\omega t + \phi - 2\pi/3) \quad (36)$$

$$v_c(t) = \hat{V} \cos(\omega t + \phi + 2\pi/3) \quad (37)$$

Setting the rotating frame angle to $\theta(t) = \omega t$:

Steady-State DC Quantities

$$\mathbf{v}_d = \hat{V} \cos \phi = \text{constant}, \quad \mathbf{v}_q = \hat{V} \sin \phi = \text{constant}, \quad \mathbf{v}_0 = 0 \quad (38)$$

Sinusoids at 50 Hz or 60 Hz appear as completely flat DC levels to the controller!

Control Advantage: Decoupled Vector Control I

In grid-connected inverters (solar, wind, battery, HVDC):

- ▶ Instantaneous active power P and reactive power Q in the dq frame (amplitude-invariant):

$$P = \frac{3}{2}(v_d i_d + v_q i_q), \quad Q = \frac{3}{2}(v_q i_d - v_d i_q) \quad (39)$$

- ▶ If we align the d -axis with the grid voltage vector such that $v_d = \hat{V}$ and $v_q = 0$:

$$\mathbf{P} = \frac{3}{2} \mathbf{v}_d \mathbf{i}_d \propto \mathbf{i}_d, \quad \mathbf{Q} = -\frac{3}{2} \mathbf{v}_d \mathbf{i}_q \propto -\mathbf{i}_q \quad (40)$$

Control Advantage: Decoupled Vector Control II

Total Decoupling

- ▶ i_d independently controls Active Power P (or DC-link voltage).
- ▶ i_q independently controls Reactive Power Q (or grid AC voltage).
- ▶ Because i_d, i_q are DC quantities in steady state, simple **Proportional-Integral (PI) controllers** achieve zero steady-state tracking error without phase lag!

Behavior Under Grid Unbalance: The 2ω Ripple I

What happens if the grid voltages are unbalanced? Recall that unbalanced voltages contain both positive ($\bar{V}_1$) and negative ($\bar{V}_2$) sequences:

$$\vec{v}_{\alpha\beta}(t) = \hat{V}_1 e^{j(\omega t + \phi_1)} + \hat{V}_2 e^{-j(\omega t - \phi_2)} \quad (41)$$

Rotating with the forward angle $\theta = \omega t$:

$$v_{dq}(t) = \vec{v}_{\alpha\beta}(t) e^{-j\omega t} = \underbrace{\hat{V}_1 e^{j\phi_1}}_{\text{DC component}} + \underbrace{\hat{V}_2 e^{-j(2\omega t - \phi_2)}}_{\text{Double-frequency AC ripple at } 2\omega} \quad (42)$$

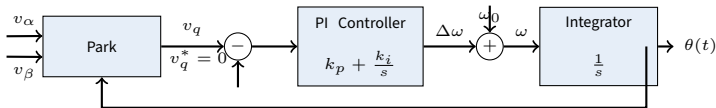
Behavior Under Grid Unbalance: The 2ω Ripple II

Key Diagnostic Insight

- ▶ Positive sequence appears as a pure **DC level** in the dq frame.
- ▶ Negative sequence appears as a **100 Hz (or 120 Hz) sinusoidal oscillation** in v_d and v_q !
- ▶ Dual-frame controllers (DDSRF-PLL) separate sequences to ride through asymmetric faults.

Grid Synchronization: The SRF-PLL I

To execute the Park transform, we must track the exact grid voltage angle $\theta(t)$ in real time. This is achieved by the **Synchronous Reference Frame Phase-Locked Loop (SRF-PLL)**:



Operating Principle: If the estimated angle θ matches the grid angle, $v_q = 0$. The PI controller adjusts estimated frequency ω until error $v_q \rightarrow 0$.

Bridges and Connections Between Transforms

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Comparing Fortescue, Clarke, and Park

Feature	Fortescue (012)	Clarke ($\alpha\beta 0$)	Park ($dq0$)
Domain	Complex Phasor (Frequency)	Instantaneous Time	Instantaneous Time
Coordinates	Complex scalars $\bar{V}_0, \bar{V}_1, \bar{V}_2$	Stationary orthogonal ($\alpha, \beta, 0$)	Synchronous rotating ($d, q, 0$)
Balanced Steady State	$\bar{V}_1 = V_{rms}\angle\phi, \bar{V}_2 = 0, \bar{V}_0 = 0$	Rotating circle of radius $\hat{V}$ at ω	Constant DC values ($V_d, V_q = \text{const}$)
Unbalanced Steady State	Non-zero $\bar{V}_2$ and $\bar{V}_0$ (constant phasors)	Ellipse on $\alpha\beta$ plane	DC value + 2ω AC ripple in d and q
Primary Strengths	Frequency-domain fault calculation, symmetrical networks, protection	Space Vector PWM, instant power calculation, converter modeling	Linear PI control, machine flux/torque decoupling, PLL synchronization

Table 2: Unified comparison of coordinate transformations.

Mapping Between Domains I

Under fundamental frequency operation, exact mappings link these three representations:

Mapping Between Domains II

Clarke $\longleftrightarrow$ Fortescue

The complex space vector relates directly to positive and negative sequence phasors:

$$\vec{v}_{\alpha\beta}(t) = \sqrt{2} (\bar{V}_1 e^{j\omega t} + \bar{V}_2^* e^{-j\omega t}) \quad (43)$$

- ▶ Forward rotating part $\rightarrow \bar{V}_1$
- ▶ Backward rotating part $\rightarrow \bar{V}_2$
- ▶ Homopolar part $v_0(t) = \sqrt{2} \operatorname{Re}\{\bar{V}_0 e^{j\omega t}\}$

Mapping Between Domains III

Park $\longleftrightarrow$ Fortescue

Demodulating in the rotating frame with $\theta = \omega t$:

$$v_d(t) + jv_q(t) = \vec{v}_{\alpha\beta}(t)e^{-j\omega t} = \sqrt{2}\bar{V}_1 + \sqrt{2}\bar{V}_2^*e^{-j2\omega t} \quad (44)$$

- ▶ Mean (DC) value gives **positive sequence**!
- ▶ Amplitude of 2ω ripple gives **negative sequence**!

Kron Reduction (Network Admittance Reduction)

Kron Reduction: Motivation I

In power system dynamic studies and transient stability simulations:

- ▶ Bulk power grids contain tens of thousands of transmission and distribution buses.
- ▶ However, dynamic states (differential equations) reside almost exclusively in **synchronous machines** and **converter-interfaced generation**.
- ▶ Most intermediate transmission and substation buses have **zero current injection** ($I_{\text{bus}} = 0$).
- ▶ Passive loads can often be converted to equivalent constant impedances $Y_L = \frac{P_L - jQ_L}{|V|^2}$ at nominal voltage.

Kron Reduction: Motivation II

The Question Gabriel Kron Answered (1939): Can we algebraically eliminate all non-generator / passive buses while preserving the **exact multiport electrical relationships** between generator internal voltages?

Mathematical Derivation: The Schur Complement I

Partition the nodal admittance equations $\mathbf{I} = \mathbf{Y}_{\text{bus}} \mathbf{V}$ into:

- ▶ Subset 1 (or G): **Kept buses** (e.g., generator internal EMF nodes).
- ▶ Subset 2 (or L): **Eliminated buses** (zero-injection nodes, $\mathbf{I}_2 = \mathbf{0}$).

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_{11} & \mathbf{Y}_{12} \\ \mathbf{Y}_{21} & \mathbf{Y}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} \quad (45)$$

From the second block row:

$$\mathbf{0} = \mathbf{Y}_{21} \mathbf{V}_1 + \mathbf{Y}_{22} \mathbf{V}_2 \quad (46)$$

Mathematical Derivation: The Schur Complement II

Assuming $\mathbf{Y}_{22}$ is invertible (requires at least one shunt path to ground in the eliminated network):

$$\mathbf{V}_2 = -\mathbf{Y}_{22}^{-1} \mathbf{Y}_{21} \mathbf{V}_1 \quad (47)$$

Substituting $\mathbf{V}_2$ into the first block row:

$$\mathbf{I}_1 = \mathbf{Y}_{11} \mathbf{V}_1 + \mathbf{Y}_{12} (-\mathbf{Y}_{22}^{-1} \mathbf{Y}_{21} \mathbf{V}_1) \quad (48)$$

Factoring out $\mathbf{V}_1$:

$$\mathbf{I}_1 = \underbrace{(\mathbf{Y}_{11} - \mathbf{Y}_{12} \mathbf{Y}_{22}^{-1} \mathbf{Y}_{21})}_{\mathbf{Y}_{\text{Kron}}} \mathbf{V}_1 \quad (49)$$

Mathematical Derivation: The Schur Complement III

Kron Reduced Admittance Matrix

$$\mathbf{Y}_{\text{Kron}} = \mathbf{Y}_{11} - \mathbf{Y}_{12}\mathbf{Y}_{22}^{-1}\mathbf{Y}_{21} \quad (50)$$

This is mathematically the **Schur complement** of block $\mathbf{Y}_{22}$ in $\mathbf{Y}_{\text{bus}}$.

Properties and Applications of Kron Reduction I

1. Exact Multiport Equivalence:

- ▶ The terminal behavior seen from the kept buses is **100% mathematically exact**.
- ▶ No dynamic or steady-state approximation is made (given linear load impedances).

2. Graph-Theoretic Effect: Fill-In:

- ▶ While original power grid admittance matrices are extremely **sparse** ($> 99\%$ zeros), Kron reduction creates equivalent direct transfer admittances between all buses connected through the eliminated network.
- ▶ $\mathbf{Y}_{\text{Kron}}$ becomes a **dense** matrix!

Properties and Applications of Kron Reduction II

- ▶ This corresponds to generalized Star-Delta (Y- Δ) transformations.

3. Classical Transient Stability Simulation:

- ▶ In electromechanical stability (the swing equation):

$$M_i \frac{d^2 \delta_i}{dt^2} = P_{m,i} - P_{e,i} \quad (51)$$

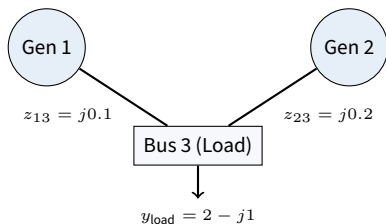
- ▶ Electrical power output of generator i can be written in closed-form:

$$P_{e,i} = E_i^2 G_{ii} + \sum_{j \neq i} E_i E_j (G_{ij} \cos(\delta_i - \delta_j) + B_{ij} \sin(\delta_i - \delta_j)) \quad (52)$$

where $Y_{ij} = G_{ij} + jB_{ij}$ are the elements of the **Kron-reduced admittance matrix**!

Example: 3-Bus System Reduced to 2 Generator Buses I

Consider two generators (Buses 1 and 2) feeding a common load (Bus 3) with line impedances $z_{13} = j0.1$, $z_{23} = j0.2$, and a load admittance $y_{\text{load}} = 2 - j1$ at Bus 3:



Example: 3-Bus System Reduced to 2 Generator Buses II

Admittances: $y_{13} = \frac{1}{j0.1} = -j10$, $y_{23} = \frac{1}{j0.2} = -j5$. The 3×3 admittance matrix is:

$$\mathbf{Y}_{\text{bus}} = \begin{bmatrix} -j10 & 0 & j10 \\ 0 & -j5 & j5 \\ j10 & j5 & (2 - j16) \end{bmatrix} \quad (53)$$

Eliminating Bus 3: $Y_{11} = \begin{bmatrix} -j10 & 0 \\ 0 & -j5 \end{bmatrix}$, $Y_{12} = \begin{bmatrix} j10 \\ j5 \end{bmatrix}$, $Y_{21} = \begin{bmatrix} j10 & j5 \end{bmatrix}$, $Y_{22} = [2 - j16]$.

Example: 3-Bus System Reduced to 2 Generator Buses III

$$\mathbf{Y}_{\text{Kron}} = Y_{11} - Y_{12}Y_{22}^{-1}Y_{21} \quad (54)$$

$$= \begin{bmatrix} -j10 & 0 \\ 0 & -j5 \end{bmatrix} - \frac{1}{2 - j16} \begin{bmatrix} -100 & -50 \\ -50 & -25 \end{bmatrix} \quad (55)$$

$$= \begin{bmatrix} 0.769 - j3.846 & 0.385 + j3.077 \\ 0.385 + j3.077 & 0.192 - j3.462 \end{bmatrix} \quad (56)$$

Insight: Gen 1 and Gen 2 are now directly coupled by the transfer admittance

$Y_{12}^{\text{Kron}} = 0.385 + j3.077$. Bus 3 has been completely absorbed!

Discrete Fourier Transform (DFT) for Phasor Estimation

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The Phasor Estimation Problem I

In the real world:

- ▶ Physical voltage and current waveforms are continuous analog signals $x(t)$.
- ▶ Digital relays, disturbance fault recorders (DFRs), and Phasor Measurement Units (PMUs) sample the waveforms using Analog-to-Digital Converters (ADCs) at sampling frequency $f_s = N \cdot f_0$ (e.g., $N = 64$ samples per cycle).
- ▶ Power system analysis algorithms (symmetrical components, distance protection, power flow, state estimation) require **complex phasors** $\bar{X} = X_{\text{rms}}e^{j\phi}$.

How do we transform discrete sampled time values into accurate fundamental frequency phasors? $\implies$ The Full-Cycle Discrete Fourier Transform (DFT) filter.

The Full-Cycle DFT Filter Formulation I

Let $x[k] = x(k\Delta t)$ be the sequence of sampled values. For a window of N samples over one fundamental cycle:

$$\bar{X} = \frac{\sqrt{2}}{N} \sum_{k=0}^{N-1} x[k] e^{-j \frac{2\pi k}{N}} = \frac{\sqrt{2}}{N} \sum_{k=0}^{N-1} x[k] \left[\cos \left(\frac{2\pi k}{N} \right) - j \sin \left(\frac{2\pi k}{N} \right) \right] \quad (57)$$

Real Part (Cosine Filter)

$$X_{\text{Re}} = \frac{\sqrt{2}}{N} \sum_{k=0}^{N-1} x[k] \cos \left(\frac{2\pi k}{N} \right)$$

Imaginary Part (Sine Filter)

$$X_{\text{Im}} = -\frac{\sqrt{2}}{N} \sum_{k=0}^{N-1} x[k] \sin \left(\frac{2\pi k}{N} \right)$$

The Full-Cycle DFT Filter Formulation II

Phasor Magnitude and Phase:

$$X_{\text{rms}} = \sqrt{X_{\text{Re}}^2 + X_{\text{Im}}^2}, \quad \phi = \text{atan2}(X_{\text{Im}}, X_{\text{Re}}) \quad (58)$$

Filtering Properties of the Full-Cycle DFT I

Complete Harmonic Rejection

Because $\sum_{k=0}^{N-1} e^{-j\frac{2\pi km}{N}} = 0$ for any integer $m \neq 0$:

- ▶ **DC offset rejection:** A constant DC component evaluates to zero.
- ▶ **Integer harmonics:** 2nd (100 Hz), 3rd (150 Hz), 5th (250 Hz), etc., are **completely eliminated!**

Filtering Properties of the Full-Cycle DFT II

Limitations in Practice

- ▶ **Decaying DC transients:** Fault currents contain exponentially decaying DC ($I_{dc}e^{-t/\tau}$). A standard DFT filter leaks some decaying DC into the estimate $\implies$ requires a Mimic Filter ($1 + \tau \frac{d}{dt}$).
- ▶ **Off-nominal frequency:** If grid frequency drifts to 49.8 Hz, spectral leakage occurs $\implies$ requires frequency tracking / interpolation.

Summary and Conclusions

The background of the slide features a white upper half and a teal lower half. The teal section is composed of two large triangular shapes that meet at a point at the bottom center, creating a V-shape. The teal color is a dark, muted blue-green.

Master Summary of Power System Transforms

Transform	Coordinates	Mathematical Nature	Key Engineering Benefit
Fortescue (1918)	Phasor (0, 1, 2)	Eigenvector decomposition of cyclic matrices	Decoupled positive, negative, and zero sequence networks; fault calculation; unbalance protection (46, 51N).
Clarke (1938)	Stationary ($\alpha, \beta, 0$)	3-phase to 2-axis orthogonal projection	Space Vector modulation (SVPWM); instant power $p-q$; fast fault detection.
Park (1929)	Synchronous ($d, q, 0$)	Angle-dependent 2D rotation $\mathbf{R}(\theta)$	Eliminates rotor position from machine equations; converts AC into DC; decoupled $P-Q$ control; SRF-PLL.
Kron Reduction (1939)	Admittance matrix $\mathbf{Y}_{bus}$	Schur complement: $\mathbf{Y}_{11} - \mathbf{Y}_{12} \mathbf{Y}_{22}^{-1} \mathbf{Y}_{21}$	Eliminates interior passive nodes; essential for transient stability simulations and dynamic equivalencing.
Full-Cycle DFT	Sampled time to phasor	Discrete orthogonal trigonometric filter	Extracts RMS and phase angle from ADC samples; rejects DC and harmonics; standard in PMUs and numerical distance relays.

Table 3: Master reference table of power system transforms.

Key Takeaways I

1. **Decoupling is the primary motivation:** Real three-phase physical components have mutual electromagnetic couplings. Transforms find the natural eigenspace where equations become independent single-phase systems.
2. **Choosing the right domain:**
 - ▶ For steady-state unbalance and fault currents → **Fortescue Symmetrical Components**.
 - ▶ For instantaneous converter switching and trajectory visualization → **Clarke Transform**.
 - ▶ For closed-loop inverter control and rotating machinery → **Park Transform**.
 - ▶ For network simplification and multi-machine dynamics → **Kron Reduction**.
 - ▶ For real-time measurement and digital relaying → **Full-Cycle DFT**.

Key Takeaways II

3. **Cross-domain connections:** Positive sequence in Fortescue corresponds to a counter-clockwise circle in $\alpha\beta$, which collapses into a stationary DC point in the dq frame! Negative sequence creates a 100 Hz ripple in dq .

Thank You!

Questions and Discussion

Python companion code and interactive notebook available in:
`Lectures/transforms/transforms.py` and `transforms.ipynb`